GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Network Theory — Official GATE ECE Syllabus

Paper Code: EC • Section 2: Networks, Signals and Systems (8 - 11 Marks)

Weightage: 8 - 11 Marks • **Subject Code:** NT • **Question Types:** MCQ & NAT Heavy

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: CIRCUIT ANALYSIS & SOLUTION METHODS

- Network solution methods: Nodal analysis, Mesh analysis, Supernode and Supermesh methods.
- Node and mesh matrix equations: Inspection method for resistive and reactive networks with independent and dependent sources.
- Network theorems: Superposition theorem (valid only for linear systems, power calculation trap), Thevenin's theorem, Norton's theorem.
- Maximum power transfer theorem for DC networks and AC networks with reactive impedances (conjugate matching).
- Wye-Delta (Star-Delta) transformations and source transformation techniques.
- Tellegen's theorem, Reciprocity theorem, Millman's theorem, and Substitution theorem.

SECTION: SINUSOIDAL STEADY-STATE ANALYSIS & RESONANCE

- Sinusoidal steady-state analysis using phasors, impedance, and admittance concepts.
- Active power (P), reactive power (Q), apparent power (S), and complex power. Power factor and power factor correction.
- Resonance in electrical circuits: Series RLC resonance, parallel RLC resonance, anti-resonance.
- Resonance parameters: Resonant frequency, half-power frequencies, bandwidth (BW), Quality factor (Q-factor), selectivity.
- Coupled circuits: Self-inductance, mutual inductance, dot convention, coefficient of coupling (k), equivalent T and Pi models.

SECTION: TRANSIENT AND TIME-DOMAIN ANALYSIS

- Time-domain analysis of linear circuits: First-order RL and RC circuits, time constant calculation (tau = L/R, tau = RC).
- Differential equations for second-order RLC circuits: Overdamped, critically damped, and underdamped transient responses.
- Damping ratio (zeta) and undamped natural frequency (omega_n).
- Evaluation of initial conditions (t = 0-) and switching conditions (t = 0+): Inductor current continuity (i_L(0-) = i_L(0+)) and capacitor voltage continuity (v_C(0-) = v_C(0+)).
- Step, impulse, and ramp response of linear networks. Steady-state error evaluation.
- Laplace transform domain analysis of networks: Transformed s-domain equivalent models for inductors and capacitors with initial conditions.

SECTION: TWO-PORT NETWORKS & STATE EQUATIONS

- Two-port network parameter representations: Impedance parameters (Z), Admittance parameters (Y), Hybrid parameters (h), Inverse Hybrid parameters (g).
- Transmission parameters (ABCD) and Inverse Transmission parameters (A'B'C'D').

- Inter-parameter conversions and relationships.
- Symmetry and Reciprocity conditions for all two-port parameter sets (e.g., Z12 = Z21 for reciprocity, Z11 = Z22 for symmetry).
- Interconnection of two-port networks: Series-series (Z), parallel-parallel (Y), cascade (ABCD).
- State equations and matrix formulation for linear active and passive networks.

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

- Engineering Circuit Analysis — William H. Hayt & Jack E. Kemmerly (McGraw Hill)
- Network Analysis — M.E. Van Valkenburg (Prentice Hall)
- Fundamentals of Electric Circuits — Charles K. Alexander & Matthew N.O. Sadiku

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

- In Maximum Power Transfer with dependent sources, Thevenin resistance must be calculated using test source method (V_test / I_test).
- Initial conditions on inductors (open circuit at t=0+ if i(0-)=0) and capacitors (short circuit at t=0+ if v(0-)=0) frequently appear in NAT questions.
- Two-port cascade connection multiplication: ABCD_total = ABCD_1 * ABCD_2 (matrix order matters!).